

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

INDEX NUMBER

BIOLOGY

9648/03

Paper 3 Applications Paper and Planning Question

21 September 2011

2 hours

Additional Materials: Answer Paper
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

The number of marks is given in the brackets [] at the end of each question or part question.

For Examiner's Use	
1	13
2	12
3	15
4 (Planning)	12
5	20
Total	72
Percentage	
Grade	

This document consists of **15** printed pages.



Answer **all** questions

- 1 Diabetes is a condition where glucose level in the blood is not regulated normally by the body. The excessively high levels of glucose in the blood for prolonged periods may damage organs and body systems. Fig. 1.1 shows the changes in the glucose level after a meal of a normal person and a diabetic.

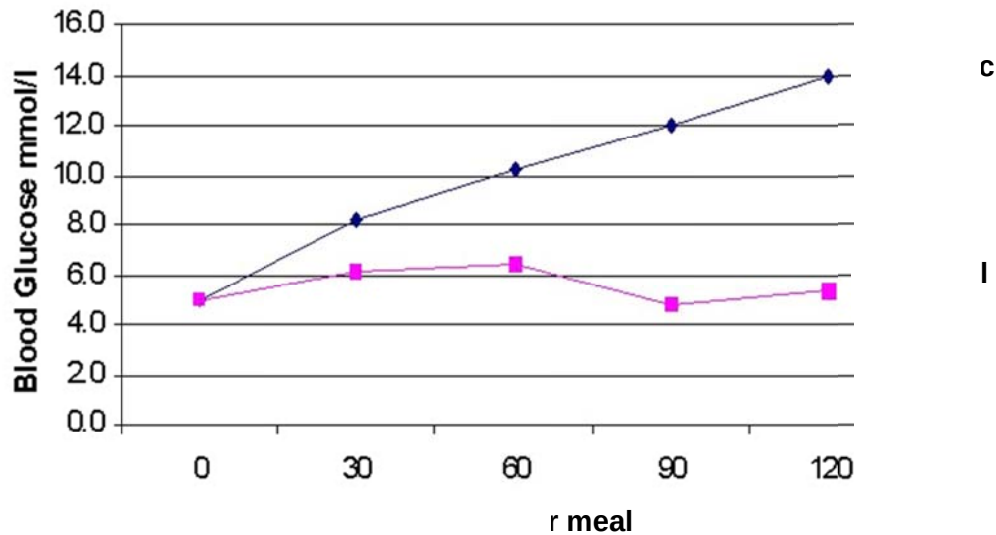


Fig. 1.1

- (a) With reference to Fig. 1.1, state and explain the differences in glucose level between the normal person and the diabetic after meal.

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[3]

(b) Describe the steps required to produce a recombinant plasmid containing the proinsulin gene.

[3]

Describe the steps involved in transformation.

[2]

The transformed bacteria were then cultured on an agar plate. The plasmid pBR322 was used for the production of recombinant plasmids. There are two antibiotic resistance genes in the plasmid, coding for ampicillin resistance and tetracycline resistance. The restriction enzyme used for the cleaving of the plasmid was *Bam*HI and would cleave at the restriction site as shown in Fig 1.2.

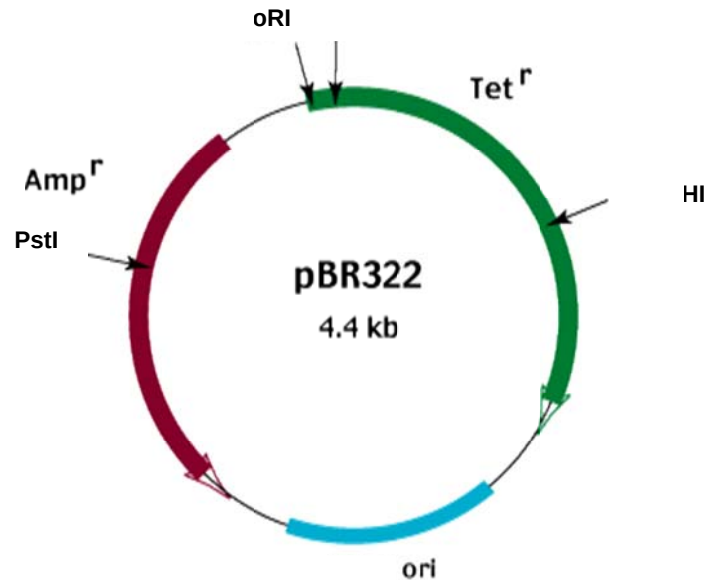


Fig 1.2

- (d) Outline the steps that will be taken to select and culture the bacteria with the recombinant plasmids.

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- (e) Explain why the gene product obtained from the bacteria cells cannot be used directly to treat diabetic patient.

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..... [2]

[Total: 13]

- 2 Sickle cell anaemia is a genetic disease in which red blood cells become sickle-shaped when the oxygen concentration is low. Fig. 2.1 shows the inheritance of the allele in a family where the condition exists.

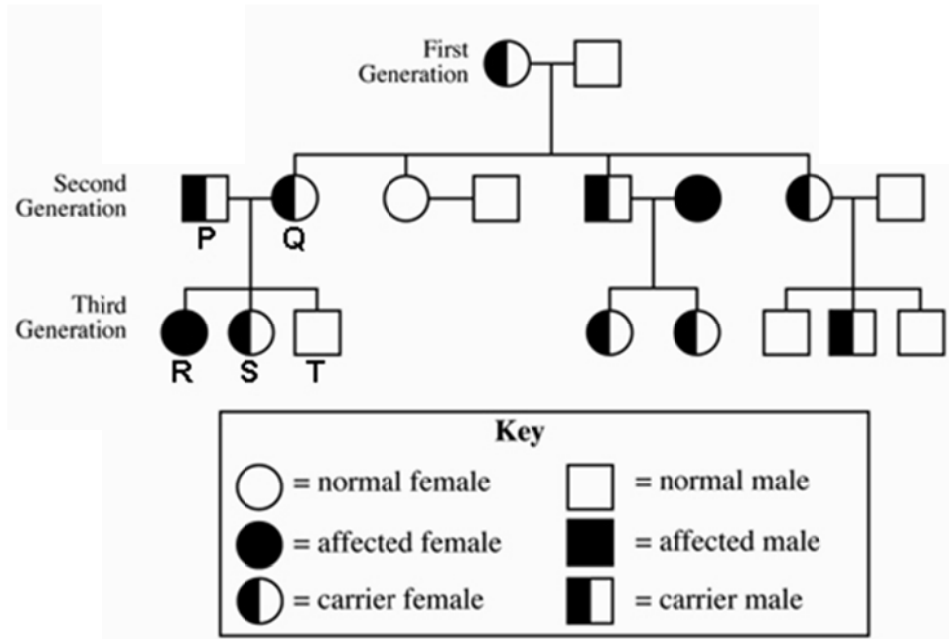


Fig. 2.1

- (a) With reference to Fig. 2.1
- (i) Explain why none of the four direct descendants of the first generation express sickle cell anaemia.
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- [2]
- (ii) Calculate the probability of **P** and **Q** having a male child who has sickle cell anaemia (show your workings).

[1]

The sickle cell allele (HbS) differs from the normal allele (HbA) by a single nucleotide, and the restriction enzyme *Mst* II can be used to differentiate between the two alleles. The positions that *Mst* II cleaves the respective alleles are shown in Fig. 2.2.

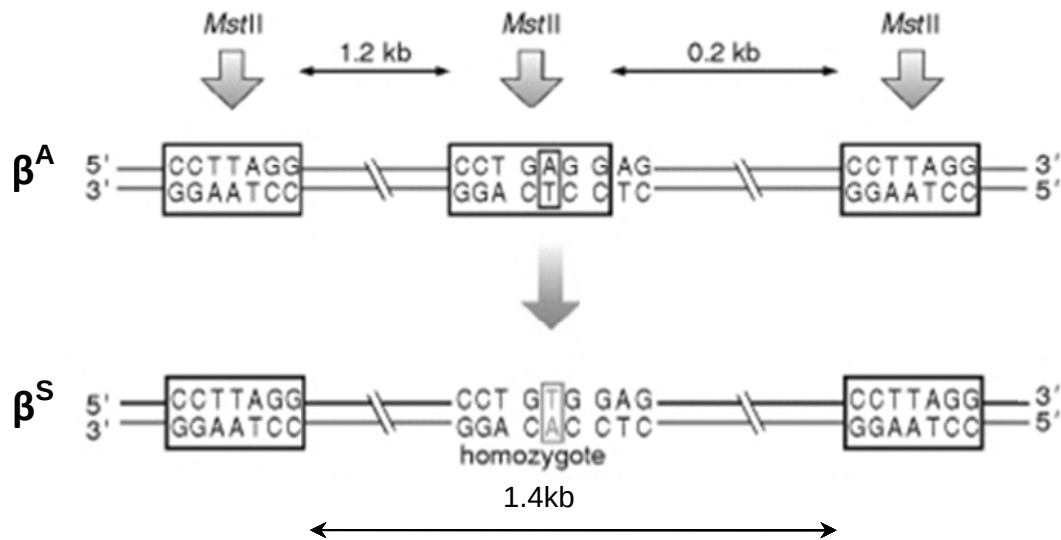


Fig. 2.2

- (b) Explain why the restriction enzyme only binds to specific restriction sites.

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- (c) Explain how *Mst* II cleaves the two alleles differently.

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After processing and digesting the DNA of the family members in Fig. 2.1 to *Mst* II, gel electrophoresis was carried out to separate the DNA fragments for analysis. Fig.2.3 shows the incomplete gel electrophoresis results from the analysis.

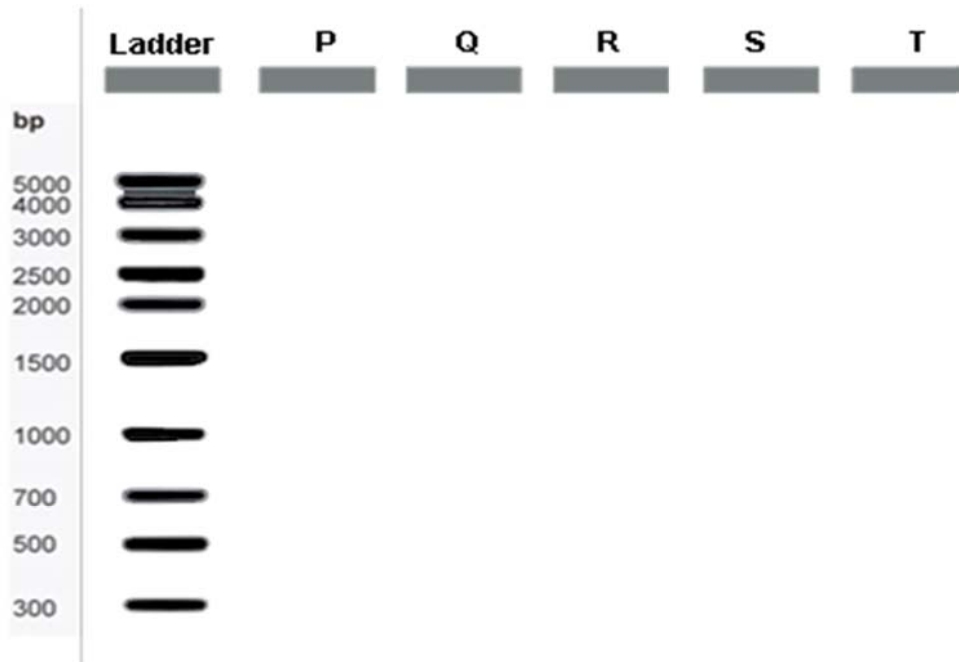


Fig. 2.3

- (d) Complete the diagram in Fig. 2.3 for individuals **P**, **Q**, **R**, **S** and **T**, by drawing bands in their lanes, with the aid of information from Fig. 2.1 and Fig. 2.2. [2]
- (e) Describe how Southern blotting (nucleic acid hybridization) could be applied after gel electrophoresis to further reduce the number of bands visualised to one band per allele.

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..... [3]

[Total: 12]

- 3** Every year, farmers in North America lose billions of dollars when their cattle get shipping fever. Shipping fever is a form of pneumonia caused by the bacterium *Mannheimia haemolytica*. The disease is often triggered by the stress of being transported and hence the name shipping fever. The disease is often associated with transporting calves from diverse geographic, nutritional, and genetic backgrounds. A variety of stress factors during transportation suppress calves' defense mechanisms resulting in respiratory tract infections.

The disease can be controlled by vaccination using a weakened strain of the bacterium. However, its usage is expensive, raises controversy and sometimes increases mortality of the cattle lot.

A protein, leukotoxin, from *M. haemolytica* is found to be responsible for the symptoms of shipping fever. Scientists developed an edible vaccine by inserting part of the leukotoxin gene juxtaposed to a green fluorescent protein (GFP) gene, into the DNA of white clover, a cattle feed, using a vector. The transgenic clovers were planted in plots with plant beds spaced 1 m apart to facilitate mandatory debudding, a precautionary measure.

The polypeptide produced, purified by gel electrophoresis, was shown to elicit toxin-neutralising antibodies when tested for immunogenicity.

- (a)(i)** Suggest how transporting cattle with '*diverse geographic, nutritional, and genetic backgrounds*' increases the chance of cattle developing shipping fever.

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[1]

- (ii)** Explain how transportation stress contributes to the development of shipping fever.

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[1]

- (b)(i)** Suggest what '*controversies*' might have arisen with the use of weaken strains of *M. haemolytica* as vaccine.

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[1]

- (ii)** Suggest why edible vaccines are less expensive to produce and use than vaccines delivered by injections.

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[1]

- (c) Suggest **two** reasons why only part and not the whole of the leukotoxin gene was inserted into the white clover.

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- (d)(i) Describe how this DNA could be inserted into white clover cells.

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..... [2]

- (ii) Explain the purpose of GFP gene.

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- (e) Describe how the modified white clover cells can be grown to give white clover plants containing part of the leukotoxin gene.

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..... [2]

- (f) Suggest why debudding is required for transgenic clovers.

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..... [1]

- (g) Describe how the leukotoxin protein is purified using gel electrophoresis.

[3]

[Total: 15]

Planning Question

- 4 Respiration begins when a seed absorbs water. The enzymes inside seeds have to be in suspension to function. Activated enzymes start respiration and use organic matter stored in the seed during respiration to make ATP molecules needed for growth. Respiration continues in the seed after the sprout has emerged until the seed's endospermic material is used up and the cotyledons have produced the sprout.

You are required to compare respiration rate of germinating green bean seedlings of different sizes.

Your planning must be based on the assumption that you have been provided with the following equipment and materials:

- Syringe
- Rubber connecting tube
- Glass capillary tube with bore diameter of 0.4 mm
- Soda lime granules
- Germinating green bean seedlings of different sizes
- Glass beads
- Coloured liquid (manometer fluid)
- Ruler / Graph paper
- Stop watch
- Weighing balance
- Paper towels

[Total:12]

Your plan should have a clear and helpful structure to include:

- an explanation of the theory to support your practical procedure
- a description of the method used, including the scientific reasoning behind the method and any recommended safety measures
- an explanation of the dependent and independent variables involved
- relevant, clearly labelled diagrams
- how you will record your results and ensure that they are as accurate and reliable as possible
- proposed layout of results table and graphs with clear headings and labels
- the correct use of technical and scientific terms

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Handwriting practice lines consisting of multiple rows of dashed lines for tracing and solid lines for writing.

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Free-response Question

Write your answers on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate.
- should be in continuous prose, where appropriate.
- must be set out in section **(a)**, **(b)** etc., as indicated in the question.

5 (a) Describe the characteristics of stem cells. [4]

(b) With reference to **two** types of stem cells, explain the normal functions of stem cells in an organism. [6]

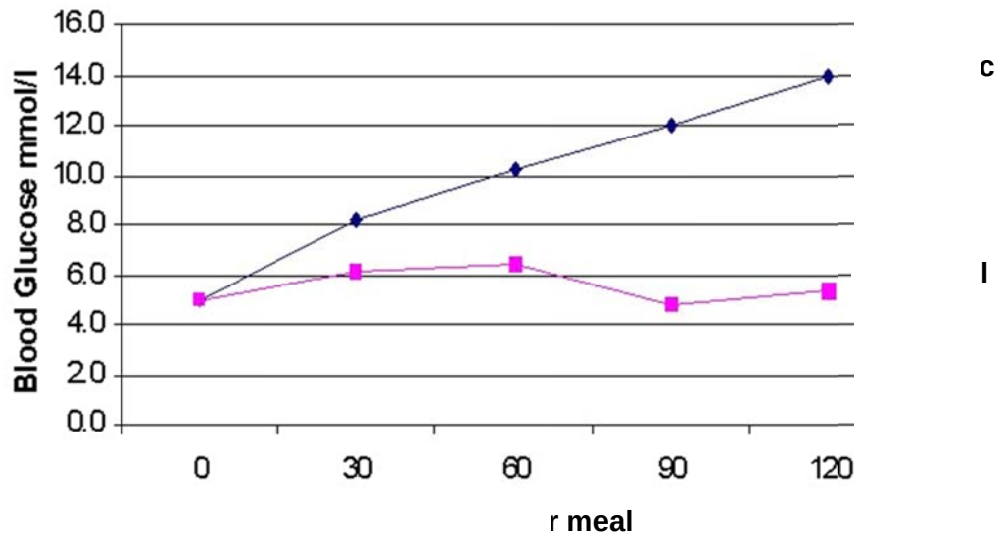
(c) Explain the factors that prevent gene therapy from becoming an effective treatment for genetic diseases. [10]

[Total: 20]

End of Paper

Suggested Mark Scheme

- 1 Diabetes is a condition where glucose level in the blood is not regulated normally by the body. The excessively high levels of glucose in the blood for prolonged periods may damage organs and body systems. Fig. 1.1 shows the changes in the glucose level after a meal of a normal person and a diabetic.

**Fig. 1.1**

- (a) With reference to Fig. 1.1, state the differences in glucose level between the normal person and the diabetic after meal.

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..... [3]

Insulin deficiency is one of the causes of diabetes. With the advances in biotechnology, genetic engineering can be used to produce large quantities of human insulin and this can be used to treat diabetics.

- (b) State precisely where, in the human body, insulin is produced.

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- (c) Describe the steps required to produce a recombinant plasmid containing the proinsulin gene.

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[4]

- (d) The recombinant insulin plasmids are inserted into bacteria host cells via transformation.

Describe the steps involved in transformation.

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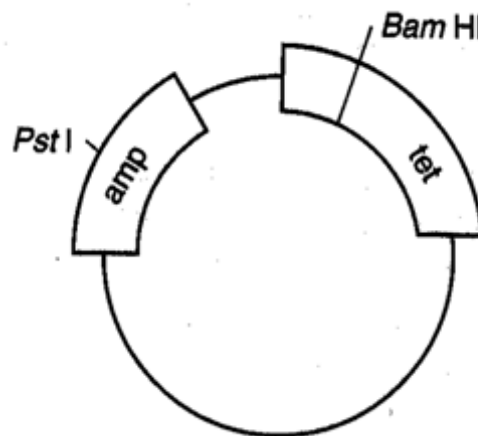
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The transformed bacteria were then cultured on an agar plate. The plasmid pBR322 was used for the production of recombinant plasmids. There are two antibiotic resistance genes in the plasmid, coding for ampicillin resistance and tetracycline resistance. The restriction enzyme used for the cleaving of the plasmid was *Bam* HI and would cleave at the restriction site as shown in Fig 1.2.



Plasmid pBR322

Fig 1.2

- (e) Outline the steps that will be taken to select and culture the bacteria with the recombinant plasmids.

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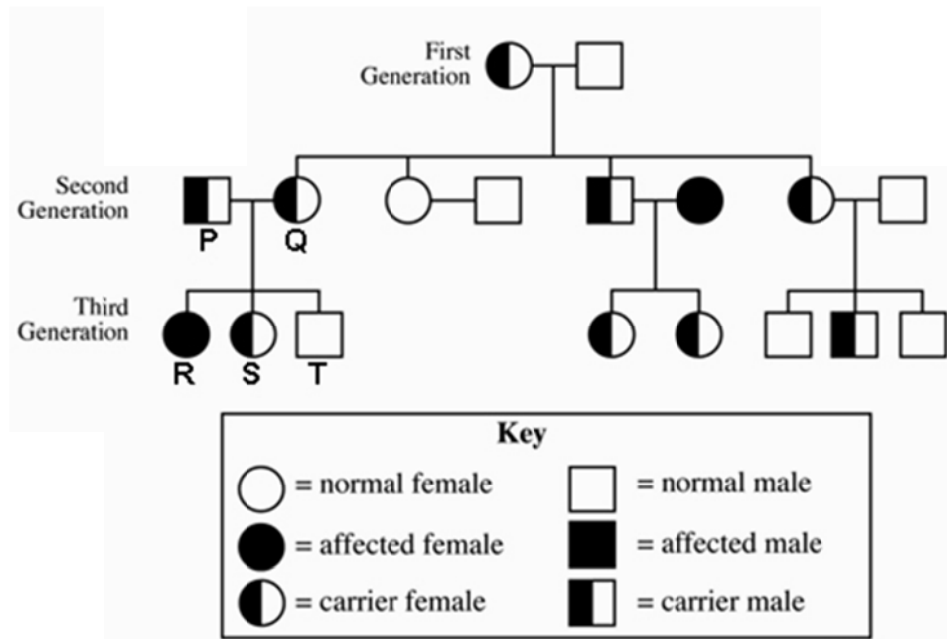


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[Turn over

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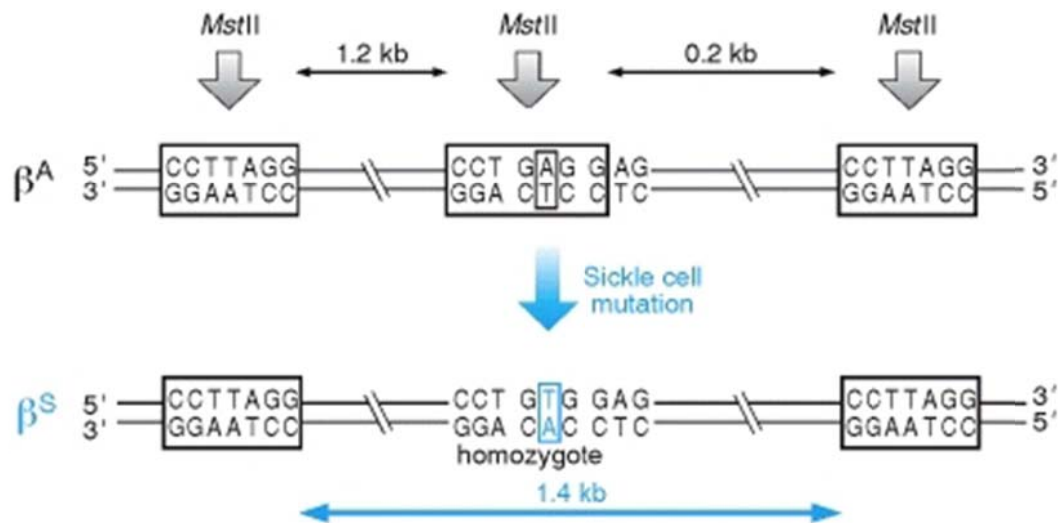


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- (b) Explain why the restriction enzyme only binds to specific restriction sites.

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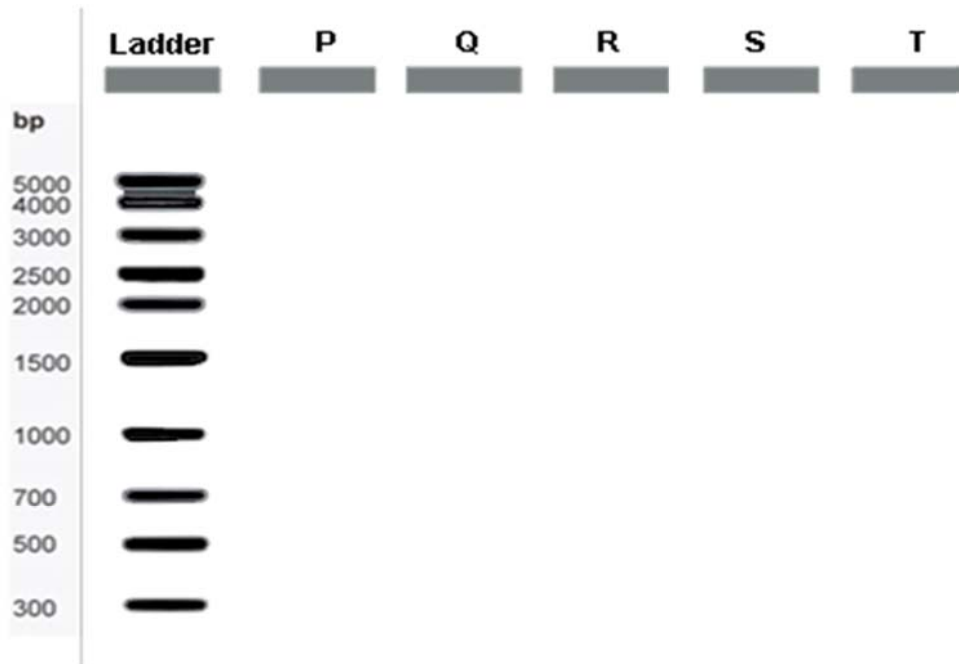


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[3]

- (f) Suggest an improvement for the DNA ladder used in the investigation.

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[1]

[Total: 13]

[Turn over

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The polypeptide produced, purified by gel electrophoresis, was shown to elicit toxin-neutralising antibodies when tested for immunogenicity.

- (a)(i) Suggest how transporting cattle with '*diverse geographic, nutritional, and genetic backgrounds*' increases the chance of cattle developing shipping fever.
calves not resistant to M. haemolytica may be infected by bact carried in resistant calves fr o/r locations / breeding grounds;

[1]

- (ii) Explain how transportation stress contributes to the development of shipping fever.

overcrowding, dehydration, climate changes;

suppress immunity, calves more susceptible to opportunistic infection by respiratory bact;

[1]

- (b)(i) Suggest what '*controversies*' might have arisen with the use of weaken strains of *M. haemolytica* as vaccine.

genetic transfers / conjugation / transformation may lead to infectious bact;

[1]

- (ii) Suggest why edible vaccines are less expensive to produce and use than vaccines delivered by injections.

eliminating the need for expensive processes such as purification, extraction, needles, syringes, sterile conditions, herding, restraint

[1]

- (c) Suggest **two** reasons why only part and not the whole of the leukotoxin gene was inserted into the white clover.

plasmid vector limits size of insert;

unstable / low expression of large polypeptide;

[2]

- (f)(i) Describe how this DNA could be inserted into white clover cells.

[2]

- (ii) Explain the purpose of GFP gene.

[1]

- (g) Describe how the modified white clover cells can be grown to give white clover plants containing part of the leukotoxin gene.

[2]

- (h) Suggest why debudding is required for transgenic clovers.

[1]

- (i) Describe how the leukotoxin protein is purified using gel electrophoresis.

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[Turn over

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[illegible]

This image shows a full page of handwriting practice paper. It features multiple sets of horizontal dashed lines spaced evenly down the page, providing a guide for letter height and placement. The background is white, and there are no other markings or text present.

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